



2026 CITRIS Aviation Prize Phase 2 Proposal

Airlink Nexus: A Vertiport Framework for UC Aerial Connectivity

Shaping the Future of Aerial Campus Connectivity

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1 Introduction

Advanced Air Mobility (AAM) has the potential to transform intercampus and regional travel by allowing for fast, flexible movement using electric vertical takeoff and landing (eVTOL) aircraft. However the viability of such systems is limited not only by vehicle capability, but also by the infrastructure and operational logic required to support them. Vertiports must be placed where they are physically feasible, economically sensible, operationally efficient, and compatible with evolving regulatory and safety requirements.

Within this broader systems challenge, infrastructure remains a critical bottleneck. Traditional vertiport development often assumes ground up construction, substantial site preparation, and new passenger facing facilities. In dense urban and campus environments, such assumptions translate into high capital cost, long development timelines, and limited deployment flexibility. This means creating a system which depends on the ability to create a network of nodes rather than a single isolated site allows for more long term flexibility.

This report evaluates candidate vertiport infrastructure strategies for a distributed airlink system. From these proposed method, parking structure retrofits emerge as a candidate particularly due to parking structures serving as cost effective distributed nodes for the airlink by reducing construction scope, avoiding land acquisition, and leveraging existing urban assets while still supporting safe and efficient eVTOL operations. At the same time, this report recognizes that infrastructure design must be validated through operational analysis. Accordingly the proposed parking garage vertiport concept is positioned not simply a viability study, but as one component of a larger Airlink design effort that includes simulation driven vertiport placement, operational flow, route planning, weather and emergency considerations, and phased deployment strategy. This document establishes the infrastructure assumptions, design logic, and economic framework that is evaluated by an in depth simulation.

2 Problem Statement

Despite rapid progress in eVTOL development, the deployment of AAM networks remains constrained by a hard balance between vehicle ambition and infrastructure practicality. Existing visions of vertiport deployment often rely on large, centralized facilities that require significant capital investment, large amounts of land acquisition or heavily modified urban sites, and extensive supporting infrastructure. While such hubs may be suitable for high volume operations in the long term they are poorly aligned with the near term need for flexible, lower risk, incrementally deployable systems. For a proposed airlink, this creates three main problems. First, **infrastructure cost** limits scalability. Conventional vertiport development can require multi-millions of dollars of investment per site due to land acquisition, site preparation, utility installation, structural work, and passenger facilities. Published planning estimates range from roughly \$2M for a small single spot vertiport on an existing building to \$3.5M–\$12M for early vertiports in leading cities, with more complex urban or hub-like facilities reaching \$20M or more [1, 2, 3]. Second, **centralized deployment logic** limits network adaptability. A system built around a small number of large hubs may concentrate operations efficiently, but it reduces resilience, constrains route diversity, and creates bottlenecks in both infrastructure and operations. In contrast, a diverse and widespread airlink concept benefits from geographically distributed nodes that can connect campuses and regional access points through multiple routing options. Third, **operational feasibility must be demonstrated, not assumed**. An simulation based validation of vertiport placement provides reassurance in a branching airlink system, low-altitude vehicle routes, weather disruption, emergencies, and sense-and-avoid requirements [4] all prove to be key factors while building out a flexible simulation. A viable proposal therefore cannot stop at identifying a physically plausible vertiport site; it must show how such sites support safe and efficient operations within a larger transportation network. This work addresses these challenges by proposing a distributed vertiport strategy based on parking structure retrofits. The objective is not to replace the need for larger airport or regional nodes, but to define a lower cost class of Airlink site that can be replicated across campus and urban environments. The report evaluates the physical feasibility, economic performance, and scaling logic of a three-pad parking-structure vertiport concept, while framing that concept as an input to a broader airlink simulation and operational design effort.

3 Concept of Operations

The proposed airlink concept is structured as a distributed network of eVTOL nodes that serve different operational roles. Within this network, parking structure retrofit vertiports function as lower cost campus and urban access points, while airports and major transportation hubs serve as higher capacity regional connectors. This distributed hierarchy is intended to improve accessibility and flexibility while reducing the dependence on a small number of high capital facilities. At the node level, the proposed parking structure vertiport supports a two to three pad operational concept. Rather than maximizing pad count, the design prioritizes orderly arrival/departure sequencing, short ground movement paths, and rapid turnover. In this concept, one pad may serve as a primary arrival surface, one as a primary departure surface, and one as a flexible overflow, charging adjacent, or contingency pad depending on demand and scheduling conditions. The exact assignment would be validated through simulation and iteration, but the design philosophy is to improve utilization through role separation rather than simple pad density to accommodate a larger number of eVTOLs. Aircraft arriving at the vertiport enter via predefined low altitude approach paths that have been set up to consider major data sources like population density (assist safety and minimize noise sources) **Figures 1 and 2**, weather, and land on the assigned arrival surface. After landing, the aircraft is either repositioned to a nearby charging/staging area or held temporarily for passenger exchange, depending on state of charge, turnaround time, and queue conditions. Departing aircraft are sequenced to minimize pad conflict and reduce interference with inbound movements. Because a two- to three-pad site has limited redundancy, throughput depends strongly on reducing unnecessary repositioning, aligning passenger movement with charging operations, and minimizing overlap between arriving and departing traffic. Passenger flow is integrated with existing parking structure circulation where possible. Elevators, stairs, and access corridors reduce the need for new vertical circulation infrastructure and help connect the vertiport to campus destinations while still providing comfort to incentive passenger use. Specific implementations can be considered case by case on passenger accessibility to the vertiport site. At the conceptual level, the proposed node includes controlled passenger loading areas, safety standoff zones, and charging adjacent staging locations. Additional amenities such as weather shelter, access control, digital wayfinding, and emergency equipment can be added with lighter construction than would be required for a full terminal build. At the network level, the operational concept depends on simulation guided route planning and mission scheduling. For this proposal, this framework will assess how parking garage vertiports support campus connectivity in conjunction with airport connections as more high volume hubs, how aircraft are routed under nominal and adverse weather conditions, and how emergency or degraded mode operations affect throughput and safety. This systems level integration is essential because the value of a low cost vertiport node depends not only on whether it can be built, but on how effectively it contributes to the broader airlink.

4 Infrastructure Design

Parking Structure Selection. This work first considers the broader infrastructure problem of how to deploy a distributed airlink network using vertiport nodes that are physically feasible, economically scalable, and operationally compatible with campus and urban environments. Candidate infrastructure strategies included traditional ground up vertiport development, airport integrated facilities, surface parking lot development, and retrofit concepts based on existing multilevel parking structures. Among these, parking structures emerged as a particularly effective solution for campus scale distributed nodes because they reduce or eliminate land acquisition requirements, provide elevated open-air operating surfaces, and leverage existing circulation, utility access, and structural frameworks. This conclusion was reached through a two-stage site-selection process. First, parking structures were identified as a promising *site class* based on system-level criteria relevant to the California Airlink, including low development cost, proximity to demand centers, compatibility with rooftop operations, likely utility access, and the ability to support low-altitude eVTOL approach and departure paths with limited disruption. Second, once parking structures were identified as the preferred infrastructure class for distributed campus nodes, specific candidate facilities at each campus were screened in more detail. To move beyond a qualitative screening approach, candidate facilities were evaluated using an Analytic Hierarchy Process (AHP) combined with weighted decision matrices [5]. The primary criteria were safety considerations, structural capacity, available surface area, accessibility, ge-

ometric layout, and electrical infrastructure. This framework intentionally prioritized safety and structural plausibility over convenience alone, reflecting the design philosophy that a useful Airlink node must first be safe and technically feasible before secondary benefits can be realized. The resulting weights assigned the greatest importance to safety ($w = 0.35$), structural capacity ($w = 0.24$), and available area ($w = 0.16$), with accessibility, geometric layout, and electrical infrastructure receiving smaller but still meaningful contributions. (see **Table 1**). Applying this framework across the participating campuses identified Lower Hearst at UC Berkeley, Core West at UC Santa Cruz, North Bowl at UC Merced, and Gateway Parking Structure at UC Davis as the top-ranked candidates within their local site sets (see **Table 2**). These results support the broader conclusion that existing parking structures are strong candidates for distributed Airlink deployment because they combine large usable footprints, open-air rooftop geometry, and existing infrastructure in a way that can reduce both cost and implementation time relative to new-build alternatives. A complete description of the AHP methodology, including pairwise comparison matrices and campus-specific decision tables, is provided in **Appendix B**.

Space Requirements and Lower Hearst Case Study. Vertiport sizing was evaluated using FAA vertiport design guidance, which defines the Touchdown and Liftoff Area (TLOF), Final Approach and Takeoff Area (FATO), and associated safety areas as the primary spatial elements governing rooftop feasibility [6]. For representative eVTOL aircraft in the size class considered in this work, these regions imply an effective operational footprint on the order of roughly 80–120 ft per pad, corresponding to approximately 5,000–10,000 ft² depending on aircraft geometry and safety allowances. These values provide a practical basis for early-stage screening of candidate rooftops. To estimate whether selected sites could accommodate these requirements, rooftop dimensions were approximated using satellite imagery and cross-validated using standard parking geometry. Typical parking stalls measure approximately 9 ft by 18 ft, with aisle widths of approximately 20–24 ft, allowing rooftop dimensions to be checked against visible parking rows and circulation bands. This method provides a reproducible first-order estimate of gross rooftop area without requiring architectural drawings, while also highlighting the distinction between total roof footprint and truly usable operational area. Applying this method to the Lower Hearst Parking Structure yielded an estimated usable rooftop area of approximately 40,000 ft². While this footprint suggests that multiple pads are geometrically possible, final layout feasibility is controlled by more than area alone. Safety buffers, pad separation, circulation, terminal access, and support functions all compete for rooftop space, making operational organization as important as raw geometry. For this reason, Lower Hearst was used as the representative case study for comparing two-pad and three-pad rooftop layouts rather than simply maximizing pad count from area alone. A representative two-pad configuration for the Lower Hearst structure is shown in **Figure 3**. This layout demonstrates how FAA-informed pad geometry, passenger access, and rooftop support functions can be integrated within the available footprint while maintaining a compact and operationally coherent node.

Two-Pad and Three-Pad Configuration Trade Study. To evaluate how rooftop area affects vertiport performance, both two-pad and three-pad configurations were considered for the Lower Hearst Parking Structure. In both cases, the design follows a pad-centric operational model in which each pad supports the full cycle of landing, passenger exchange, charging, and departure. This avoids the need for aircraft repositioning across the rooftop, but increases the effective space required per pad because operational and support functions must be accommodated locally. The two-pad configuration prioritizes integration and simplicity. With two fully functional pads, the rooftop can also accommodate a compact terminal, passenger circulation, and support spaces on the same level. This reduces vertical movement, simplifies passenger flow, and supports a more compact and operationally efficient layout. By contrast, the three-pad configuration increases theoretical throughput by adding a third operational position, but does so at the cost of reduced space for supporting infrastructure. In this case, portions of the terminal, staff space, or other support functions may need to be relocated to lower levels of the structure, increasing circulation demands and operational complexity. This comparison highlights a fundamental tradeoff between capacity and integration. The three-pad concept offers higher throughput potential, but it is more sensitive to spatial constraints, congestion, and coordination requirements. The two-pad concept offers lower peak capacity, but provides a more fully integrated rooftop node with simpler circulation and potentially more reliable operation. As a result, the preferred configuration depends not only on pad count, but on whether the design objective emphasizes maximum throughput or more compact and robust operation. These differences motivate the preliminary operational model presented in the next subsection, which estimates throughput

and utilization for each configuration.

Infrastructure Design Constraints. Although parking structures are promising Airlink nodes, their adaptation as vertiports is constrained by several interacting factors. Structurally, parking decks are designed primarily for static vehicular loading and may require localized reinforcement to accommodate the concentrated and repeated dynamic loads associated with eVTOL operations [6]. A preliminary reasonableness check using the Joby aircraft as a reference case suggests that rooftop eVTOL loading is not inherently incompatible with reinforced parking structures: Joby reports a maximum gross weight of approximately 2,400 kg (5,300 lb) [7], which corresponds to average bearing pressures on the order of 35–55 psf when distributed across a representative landing gear footprint. However, this simplified estimate does not capture dynamic landing effects, fatigue, vibration, or local load-path behavior, and therefore does not remove the need for detailed structural review. Geometrically, ramps, parapets, rooftop equipment, and circulation features reduce contiguous usable area and directly constrain the number of fully operational pads that can be supported. Electrically, existing garages may provide grid access or EV charging infrastructure, but rapid-turnaround eVTOL operations may still require transformer upgrades, new distribution equipment, and site-level interconnection work [8]. Operationally, small vertiports have limited slack, so throughput is highly sensitive to turnaround time, scheduling, and queueing effects. Finally, safety and regulatory requirements related to weather, emergency procedures, low-altitude routing, and sense-and-avoid capability must also be satisfied for any candidate site [4]. Taken together, these constraints show that parking structures should be treated not as automatically viable vertiports, but as candidate infrastructure options whose feasibility depends on the combined interaction of structural capacity, spatial limits, operations, and safety.

Preliminary Vertiport Operational Model. A simplified operational model was developed to estimate vertiport throughput, pad utilization, and congestion effects for the proposed two-pad and three-pad configurations. The model assumes a pad-centric operating strategy in which each pad supports the full cycle of landing, passenger exchange, charging, and departure. The total turnaround time per aircraft is approximated as

$$t_{\text{turn}} = t_{\text{land}} + t_{\text{pax}} + t_{\text{charge}} + t_{\text{prep}} \quad (1)$$

where representative values of 2 minutes for landing and pad clearance, 3 minutes for passenger exchange, 5–8 minutes for partial fast charging, and 2 minutes for pre-departure preparation yield

$$t_{\text{turn}} \approx 12\text{--}15 \text{ minutes.} \quad (2)$$

Using this estimate, the maximum theoretical throughput per pad is

$$Q_{\text{pad}} = \frac{60}{t_{\text{turn}}} \quad (3)$$

which gives approximately 5 flights per hour per pad for a representative 12-minute turnaround time. This implies a theoretical capacity of approximately 10 flights per hour for a two-pad configuration and 15 flights per hour for a three-pad configuration. In practice, actual throughput is lower due to scheduling inefficiencies, variability in arrivals, and operational disruptions. Pad utilization can be approximated as

$$U = \frac{\lambda}{N \cdot Q_{\text{pad}}}, \quad (4)$$

where λ is the arrival rate and N is the number of pads. To avoid rapid growth in queueing delay, practical utilization is assumed to remain near 70–80% of theoretical capacity, giving an effective throughput of approximately 7–8 flights per hour for the two-pad concept and 11–12 flights per hour for the three-pad concept. These estimates reinforce the tradeoff identified in the layout study. The three-pad configuration offers higher throughput, but it is also more sensitive to congestion and coordination complexity. The two-pad configuration provides lower peak capacity, but may support more stable and integrated operation within the available rooftop footprint.

5 Economic Analysis

Capital Expenditures (CAPEX). A primary motivation for the parking-structure retrofit concept is capital efficiency. Traditional vertiport development often requires land acquisition, site preparation, new structural platforms, utility installation, and passenger-facing facilities, which can place total project cost in the multi-million-dollar range [1, 2, 3]. Such investments may be justified for a limited number of flagship hubs, but they scale poorly for a distributed Airlink network. By contrast, the retrofit concept reuses major elements of existing infrastructure and thereby reduces both construction scope and upfront cost. In the present model, a two- to three-pad parking-structure vertiport is estimated to cost approximately \$1.5M–\$2.0M, with \$1.85M used as a representative nominal value. This places the concept below many benchmark traditional vertiport estimates and supports the use of parking garages as a lower-cost class of distributed node. A detailed breakdown of assumptions, sensitivity analysis, and supporting calculations is provided in **Appendix B**.

The modeled capital cost is driven by four primary categories: structural reinforcement, pad and safety-area construction, charging infrastructure, and integration/support systems. Structural reinforcement is estimated at approximately \$150k for targeted deck strengthening, while pad construction and safety systems contribute approximately \$450k–\$750k depending on configuration and rooftop modifications required. Charging infrastructure is estimated at approximately \$600k and includes electrical upgrades, power distribution equipment, and charging hardware sized for multi-pad operation [8]. The remaining cost is attributed to integration and support systems, including access control, passenger circulation improvements, and site adaptation needed to convert an existing parking structure into an operational vertiport node. A representative cost distribution is shown in **Figure 4**. The economic significance of this structure is not only that it lowers per-site cost, but that it changes the scaling logic of the network. If distributed nodes can be deployed without ground-up land development, the Airlink can expand through a larger number of moderate-cost access points rather than a small number of very expensive hubs.

Operating Costs (OPEX). Operating costs for the proposed vertiport system are divided into infrastructure-level costs associated with the vertiport itself and system-level costs associated with aircraft operations. Annual infrastructure operating costs for a three-pad retrofit vertiport are estimated to be on the order of \$150k–\$300k. These costs are driven primarily by maintenance, staffing, utilities, and regulatory compliance. Maintenance and inspection costs include periodic structural inspection, surface upkeep, and safety system maintenance required for rooftop aviation operations. Staffing represents one of the largest contributors to infrastructure-level OPEX, with estimated costs of approximately \$75k–\$200k annually depending on staffing model and level of automation. Early-stage AAM deployments are expected to rely more heavily on personnel for monitoring, passenger handling, and safety oversight, increasing operational cost [1].

Utilities and electrical costs are driven in part by lighting, communications, and baseline infrastructure usage, but are dominated by charging infrastructure integration. While charging energy is primarily attributed to aircraft operations, supporting electrical systems and standby loads contribute to infrastructure cost [8]. Insurance and regulatory compliance further contribute to recurring costs, particularly during early deployment phases where uncertainty and risk margins are higher. Additional costs not explicitly captured in baseline estimates include security systems, access control, and monitoring infrastructure required for safe operation of rooftop vertiports. Furthermore, conversion of parking structures results in loss of revenue-generating parking spaces. Based on typical urban parking rates of \$5–\$10 per day per space and assumed utilization however university campus lost parking can be thought of in terms of lost revenue from parking passes, thus opportunity cost of displaced parking may contribute an additional \$20k–\$80k annually depending on site location and number of affected spaces.

Aircraft and Operational Costs. System level operating costs are dominated by aircraft operations, including energy consumption, pilot compensation, and maintenance. eVTOL energy consumption is commonly estimated in the range of 150–300 kWh per flight [8]. At electricity rates of approximately \$0.10–\$0.20 per kWh, this corresponds to an energy cost of roughly \$15–\$60 per flight. Pilot and maintenance costs represent a substantial portion of total operating expenses in early-stage AAM systems. Industry analyses suggest that labor and maintenance may exceed energy costs, particularly at moderate utilization levels [1]. These factors contribute significantly to the estimated total operating cost of approximately \$125 per flight derived in this study.

Cost Structure Implications. The distinction between infrastructure and operational costs highlights an important characteristic of the proposed system: while total operating costs are dominated by aircraft operations, the vertiport itself represents a relatively small portion of overall system expense. By minimizing infrastructure-related costs through retrofit strategies, the proposed approach enables scalable deployment while limiting fixed cost growth as additional nodes are introduced.

Cost per Flight and Passenger. Using the developed cost model, total system operating cost for the three-pad concept corresponds to approximately \$125 per flight under the current assumptions. For typical passenger loading assumptions, this yields an estimated per-passenger cost of roughly \$35–\$45. These values are significant because they place the infrastructure-supported operation within the same general range as projected early-stage AAM fare assumptions. They also show that the retrofit concept does not merely reduce capital cost at the expense of impractical operating economics. Instead, the parking-garage model appears capable of supporting competitive per-passenger cost, provided that utilization and scheduling remain within plausible ranges. At the same time, these figures are highly sensitive to flight frequency, load factor, and turnaround assumptions. The final simulation framework will therefore need to connect pad utilization and routing behavior directly to cost performance, rather than treating these numbers as fixed.

Stakeholder and Operating Model Considerations. The economic performance of the proposed system depends not only on cost structure, but also on the underlying ownership and operating model. Several potential stakeholder configurations were considered, including university owned vertiports with third-party aircraft operators, airport managed facilities with usage based access fees, and fully integrated private operator models in which a single entity manages both infrastructure and flight operations (**Figure 5**). In a university partnered model, capital costs may be reduced through the use of existing infrastructure, while revenue is generated through landing fees, leasing agreements, or service partnerships. In contrast, airport based nodes may introduce additional operating costs through access fees and regulatory requirements, but provide higher throughput and regional connectivity. These differing models highlight that overall system profitability is sensitive to how costs and revenues are distributed across stakeholders, and that flexible ownership structures may be necessary to support scalable deployment of the California Airlink network.

6 Profitability and Financial Viability

The profitability of the proposed concept was evaluated by comparing modeled operating cost against projected passenger revenue. Using the current per-passenger cost estimate of approximately \$35–\$45 and a representative ticket-price range of \$50–\$100, the system is capable of producing positive operating margin under moderate utilization conditions. For the current three-pad configuration, the modeled capital cost of approximately \$1.85M combined with positive annual operating margin yields a nominal payback period on the order of roughly two to three years. This result is not intended as a guarantee of near-term profitability, but as an indication that the retrofit concept is economically plausible within the market assumptions commonly cited for early-stage AAM services. Several caveats are important. First, profitability depends strongly on demand. If actual passenger uptake is weaker than assumed, the payback period extends quickly. Second, early-stage operations may face added cost from weather disruption, underutilization, staffing intensity, or conservative safety procedures. Third, airport-connected nodes may incur additional facility and access fees not present in campus-only retrofit scenarios. Even with these limitations, the financial picture remains favorable relative to more expensive traditional vertiport models. The key economic advantage of the retrofit concept is not merely high nominal return, but reduced infrastructure risk. Lower capital cost makes it easier to justify phased deployment, pilot programs, and incremental scaling while the broader California Airlink concept matures.

7 Scalability and Network Model

The parking-garage vertiport concept is most powerful when understood at the network level. Rather than attempting to replace all vertiports with one site type, this proposal treats retrofit garages as a **distributed node class** within the California Airlink. In this framework, campus and urban parking structures serve as lower-cost access points, while airports and regional hubs provide higher-capacity interfaces with broader transportation flows. This distributed architecture offers several advantages. First, it reduces reliance on a

few centralized hubs, improving resilience and route flexibility. Second, it supports *incremental deployment*: nodes can be added as demand, simulation validation, and regulatory confidence improve, rather than requiring a fully built-out system at the outset. Third, it lowers the marginal cost of expansion by using site classes that already exist in many urban and campus settings. The economics developed in this report reinforce that scaling logic. If each distributed node requires substantially less capital than a traditional vertiport, then broader geographic coverage can be achieved for the same or lower total investment. This is especially valuable in an Airlink context, where accessibility and node distribution matter as much as peak throughput at any single site.

Phased Deployment and Roadmap. The proposed parking-structure vertiport concept is designed to support a phased deployment strategy aligned with the gradual expansion of the California Airlink network. Rather than requiring large upfront capital investment, the system enables incremental rollout beginning with a small number of pilot sites at key campuses. In an initial phase, a limited network of two to four vertiports can be deployed to validate operational assumptions, including pad utilization, turnaround time, passenger flow, and integration with existing transportation systems. This phase serves as a proof-of-concept, allowing refinement of both infrastructure design and operational procedures through real-world or simulation-based feedback. Following successful validation, a second phase would expand the network by adding additional campus nodes and introducing connections to regional transportation hubs such as nearby airports. At this stage, routing complexity increases, requiring coordination between multiple vertiports, implementation of standardized operational procedures, and integration with low-altitude flight corridors. The distributed nature of the parking-structure approach enables this expansion without requiring proportional increases in capital investment, as each additional node leverages existing infrastructure rather than requiring new construction. In a final phase, the system evolves toward a fully developed California Airlink network, consisting of a combination of low-cost distributed nodes and higher-capacity hub locations as either airport vertiports or larger scale vertiports constructed for high demand regions. At this scale, resilience becomes a critical design consideration. Redundant routing between nodes, alternative landing sites, and contingency planning for weather or system disruptions are incorporated into both infrastructure placement and operational strategy. The ability to add nodes incrementally also supports adaptive scaling, allowing the network to respond to demand growth, regulatory developments, and technological advancements without requiring large-scale redesign. This phased approach reduces deployment risk by aligning infrastructure investment with demonstrated demand and operational performance. By combining incremental expansion with distributed infrastructure, the proposed system provides a practical pathway from initial pilot deployment to a resilient, statewide AAM network. However scalability is not only an economic question. It also depends on route structure, weather resilience, emergency planning, sense-and-avoid requirements, and throughput under variable demand. For that reason, the final version of this proposal will connect the distributed infrastructure concept to the simulation framework, validating not only that nodes can be built, but that they contribute effectively to safe and efficient intercampus mobility [4].

8 Safety, Weather, and SAA Considerations

Safety in the California Airlink is enforced across three layers: preflight weather screening, airspace constraint enforcement, and sense-and-avoid logic for traffic separation. These correspond to the preflight check, en-route monitoring, and conflict avoidance states in the mission state machine. The Joby S4 serves as the reference vehicle throughout; where manufacturer limits are not publicly available, conservative values with an additional safety margin have been applied.

Weather Assessment and Thresholds. Weather data is sourced from the Open-Meteo archive, sampled across a grid of points spaced approximately 0.2° by 0.25° over the NorCal region. Four parameters are retrieved at hourly resolution: wind speed, wind gust speed, visibility, and precipitation. Each grid point is classified as *unsafe* (wind ≥ 25 mph, gusts ≥ 30 mph, or visibility $< 3,000$ m), *caution* (wind ≥ 18 mph, gusts ≥ 22 mph, visibility $< 8,000$ m, or precipitation > 1.0 mm/hr), or *good*. Unsafe cells are translated into circular no-fly zones with a 9-mile radius and 15% hard buffer and removed from the routing graph entirely; caution cells generate soft-penalty zones with a 7-mile radius. The routing graph is rebuilt each hour so that route decisions always reflect current conditions.

Airspace Integration and Regulatory Compliance. Airspace data is sourced from OpenAIP in Ope-

nAIR format, covering Class B, C, D, and G boundaries as well as Prohibited and Restricted zones across the NorCal operational region. Prohibited and Restricted zones are treated as hard exclusions: any route edge intersecting these zones at the aircraft’s planned altitude is removed before search begins. Class B through G airspace receives distance-proportional soft penalties weighted by class severity (B: 2.0, C: 1.5, D: 1.0, G: 0.5), steering the optimizer toward uncontrolled corridors without prohibiting transit when no alternative exists. The modeled altitude profile uses a cruise altitude of 4,500 ft MSL with altitude-band checking to avoid false conflicts. Operations are expected to proceed under FAR Part 135, consistent with the FAA UAM ConOps v2.0 layered airspace framework.

Terrain Clearance. Terrain data is derived from USGS DEM TIF files covering the NorCal region. At 4,500 ft MSL cruise altitude the aircraft maintains substantial clearance over valley and coastal terrain on all simulated routes. Terrain is most relevant during the climb and descent phases, modeled conservatively at approximately 750 ft/min forward-flight climb rate consistent with published Joby S4 estimates.

Sense-and-Avoid. The FAA requires that aircraft operating beyond visual line of sight demonstrate an equivalent level of safety to manned aviation with respect to traffic separation. For the California Airlink, SAA is addressed at two levels. At the strategic level, fixed low-altitude corridors and population density penalty zones steer routes away from high-traffic airspace by design. At the tactical level, the state machine calls for a conflict avoidance state triggered by detected traffic or hazards, escalating to an emergency response and nearest safe landing site if separation cannot be restored. This tactical capability is implemented through three onboard sensor layers, each assigned to the flight phase and threat class for which it is best suited. *ADS-B* provides cooperative traffic awareness enroute, broadcasting and receiving position, altitude, velocity, and intent at approximately once per second. It offers no protection against noncooperative targets such as unequipped aircraft or small UAS, which motivates the next layer. *Airborne millimeter-wave radar and an IR camera* cover noncooperative detection during cruise. Radar detects aircraft-class targets at ranges of 500 to 1,500 m and maintains performance through fog and precipitation. The IR camera improves detection of low radar cross section targets such as small drones and reduces false alarms. *LiDAR* handles close-range proximity sensing from 0 to 150 m during takeoff and landing, generating centimeter-level point clouds to confirm pad clearance, detect obstacles, and support touchdown control. Its outputs directly drive the pad clearance and airspace clear for takeoff transition conditions in the pre-launch state machine. All three layers feed a unified conflict detection module, with LiDAR prioritized during ground operations, radar and IR dominating enroute, and ADS-B fused at all phases.

Emergency Landing Planning. The Airlink concept defines a tiered emergency landing strategy. The primary option is airport diversion: an emergency declaration is broadcast via ADS-B Out, granting priority handling, runway clearance, and access to ARFF services. All public-use airports within approximately 15 miles of each planned route corridor are pre-identified at mission planning time as candidate diversion sites. If no airport is reachable, an alternate network vertiport serves as a secondary fallback. As a last resort, the Joby S4’s distributed electric propulsion architecture supports a controlled low-energy descent to flat terrain, pre-catalogued from the USGS DEM data already used in the simulation, so that candidate surfaces are available to the emergency response logic without real-time terrain analysis during the contingency event.

9 Operations and Implementation

Routing Strategy. The California Airlink routing system operates as a pre-flight strategic planning process. Rather than dynamically rerouting aircraft mid-flight, all path decisions are finalized before departure, with the route validated against current airspace, weather, and population data at the time of mission request (data pulled from LandScan).

Graph Construction. The routing environment is represented as a weighted spatial graph built fresh for each departure hour. The graph overlays a uniform field grid—with nodes spaced approximately 4 miles apart—across the bounding region of the four network vertiport nodes (UC Berkeley, UC Santa Cruz, UC Merced, and UC Davis). Vertiport nodes and field nodes are connected by directed edges subject to distance constraints: field-to-field edges are limited to a 10-mile neighbor radius, airport-to-field connections extend to 15 miles, and airport-to-airport direct legs may span up to the 120-mile mission range cap. Any edge whose path intersects a hard-exclusion zone—whether a restricted airspace polygon or an unsafe weather cell—is removed from the graph before search begins.

Constraint Layers. The graph encodes three constraint types with different enforcement levels.

Hard exclusions are absolute. Restricted airspace (Class B/C/D boundaries and special-use zones) and weather cells flagged as “unsafe” are represented as circular or polygon no-fly zones with a 15% buffer factor applied. Any edge crossing one of these zones is blocked and cannot appear in any valid route.

Soft penalties are cost-based deterrents. Weather cells in “caution” status and high-population-density zones generate time penalties added to edge cost. A caution weather cell within a 7-mile radius adds a penalty equivalent to 45% of the leg’s nominal flight time; high-density population zones add 20%. Population sensitivity is time-of-day aware: the system distinguishes day (06:00–19:59) from night (20:00–05:59) and adjusts penalty zone intensity accordingly, reflecting higher noise sensitivity and safety expectations during nighttime overflights.

Routing Cost Formulation. For an edge between nodes a and b , the base flight time (in minutes) is:

$$t_{\text{flight}}(a, b) = \frac{d(a, b)}{v} \times 60 \quad (5)$$

where $d(a, b)$ is the great-circle distance in miles and $v = 120$ mph. The total edge cost is:

$$C(a, b) = t_{\text{flight}}(a, b) + P_{\text{soft}}(a, b) \quad (6)$$

The soft penalty is evaluated by sampling the edge at four interior points $\tau \in \{0.2, 0.4, 0.6, 0.8\}$ and checking proximity to each soft zone z :

$$P_{\text{soft}}(a, b) = \min \left(\sum_z \max_{\tau} p(a, b, z, \tau), 1.5 \cdot t_{\text{flight}} \right) \quad (7)$$

where the per-zone, per-sample penalty p is:

$$p(a, b, z, \tau) = \begin{cases} \alpha_z \cdot t_{\text{flight}} & \text{if } \delta_{\tau} < r_z \\ \alpha_z \cdot \phi_{\tau} \cdot \beta_z \cdot t_{\text{flight}} & \text{if } r_z \leq \delta_{\tau} < 1.2 r_z \\ 0 & \text{otherwise} \end{cases} \quad (8)$$

with:

- δ_{τ} : distance from the interpolated sample point to zone center z
- r_z : soft zone radius (scaled by factor 0.90), varying by zone type: 7 miles for weather caution, 3 miles for high-density population, and 4 miles for very-high-density population
- α_z : inner penalty coefficient, equal to 0.45 for weather zones and 0.20 for population zones
- β_z : outer band coefficient, equal to 0.20 for weather zones and 0.10 for population zones
- $\phi_{\tau} = \frac{1.2 r_z - \delta_{\tau}}{0.2 r_z} \in [0, 1]$: a linear proximity falloff in the outer buffer band

A turn penalty is applied per search step based on the heading change θ between consecutive edges:

$$\Delta_{\text{turn}} = \begin{cases} 0.15 & \text{if } \theta > 90^\circ \\ 0.05 & \text{if } 45^\circ < \theta \leq 90^\circ \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

The Dijkstra search then minimizes the total accumulated cost over a path $\{e_1, e_2, \dots, e_n\}$:

$$\text{minimize } \sum_{i=1}^n [C(e_i) + \Delta_{\text{turn}}(e_{i-1}, e_i)] \quad (10)$$

Path Search. With the graph constructed, the router executes a Dijkstra-based shortest-path search using accumulated cost—in units of effective flight minutes at 120 mph—as the optimization objective. The search first attempts a direct field-graph route between origin and destination without passing through any intermediate vertiport node. If no valid path is found, a second pass allows transfers through intermediate network nodes, with a 30-minute ground transfer penalty added at each intermediate stop to reflect realistic passenger exchange and charging time. The result is the lowest-cost route that respects all hard constraints and minimizes exposure to soft-penalty zones.

Preflight Validation and Mission Acceptance. Every mission request passes through a structured preflight check before a route is committed. As illustrated in the state machine diagram **Figures 6–9**, this process validates three conditions: aircraft readiness (battery state, maintenance status), route feasibility (a valid graph path exists within the 120-mile range cap), and weather acceptability along the planned corridor. If all three checks pass, the mission proceeds to launch clearance. If the direct route is blocked, the system automatically attempts an alternative route through the field graph before determining whether the mission must be rejected. Missions are only rejected when no safe routing option exists; a design choice that maximizes network availability while preserving safety as a non-negotiable constraint.

Emergency Landing Site Identification. A key operational safety feature of the Airlink is the identification of emergency diversion sites along every planned route. For each accepted mission, the system flags all public-use airports within the vicinity of the planned flight corridor as designated emergency landing options. This approach is consistent with standard aviation practice, under which pilots in command may divert to any available airport in the event of an emergency.

In practice, any airport within approximately 15 miles of the planned route corridor is considered reachable without significant heading deviation and is listed in the mission brief as a viable emergency destination. This leverages the same airport node connectivity already encoded in the routing graph, making emergency site identification a near-zero-cost byproduct of the planning process rather than a separate analysis step.

Simulation Implementation. The routing logic described above is implemented in a custom Python simulation framework. Graph construction, Dijkstra search, soft-penalty evaluation, and polyline simplification are all executed at mission planning time, with results passed to a front-end interface for visualization and operator review. The graph is cached by hour to avoid redundant recomputation across mission requests within the same departure window, while still refreshing when weather conditions change between hours.

Within the simulation, the four UC campus vertiport nodes serve as the fixed terminal points of the network. Demand scenarios drive mission requests between node pairs, and the router produces a complete route plan which includes departure time, estimated arrival, total distance, number of legs, and a compressed waypoint list for each accepted mission. This output feeds directly into pad-utilization and throughput analyses, connecting the routing layer to the broader operational performance evaluation of the Airlink.

The current implementation models strategic pre-flight routing only; in-flight replanning is not simulated. However, the state machine design supports conceptual in-flight weather replanning and conflict avoidance as future extensions, with the preflight route serving as the baseline from which any airborne replanning would depart.

10 Simulation Results

The routing and scheduling simulation was run across a seven-day period from February 4 through February 10, 2024, using real weather data sourced from the Open-Meteo archive for the NorCal operational region **Figure 10**. Two vertiport configurations were evaluated in parallel for each day: a two-pad layout with one VTOL assigned to each UC campus node, and a three-pad layout with two VTOLs per UC campus node. Airport nodes (KCVH, KLVK, KNUQ, KOAR, KSNS, KSQL) retained four pads and three VTOLs in both configurations. All runs used a ticket price of \$100 per passenger, autonomous operation with no pilot cost, and a minimum battery reserve of 20% at destination.

Cost Model. Operating costs in the simulation are divided into two categories. Variable costs scale with flight activity and include energy consumption, per-flight maintenance reserves, and infrastructure usage. Fixed daily costs represent ground crew wages, charger operation and maintenance, insurance, utilities, and facility overhead, and are applied to every site every operating day regardless of flight count. Fixed daily

costs are derived from the project economic model and set at \$593 per airport site and \$380–\$440 per UC campus site depending on configuration. This structure ensures that days with few or no flights still register realistic losses reflecting the staffing commitments of a 6am–10pm operating window.

Weather Impact. February 4 was a storm day with wind speeds and visibility conditions triggering the simulation’s unsafe weather thresholds across the network. Both configurations completed fewer than ten flights for the day, with over 1,250 mission requests cancelled due to weather-blocked routes. Under the revised cost model, the network recorded a daily loss of approximately \$5,100 (three-pad) and \$4,900 (two-pad), reflecting primarily the fixed daily cost of keeping ten sites staffed and operational during a weather closure. This result is substantially more realistic than earlier model versions, which produced near-zero losses on storm days by omitting fixed cost obligations.

February 7 exhibited a partial weather effect. Both configurations completed 288 to 308 flights, with cancellations spiking to 358 and 458 respectively, compared to a typical clear-day range of 138 to 178. Daily profit fell to approximately \$39,600–\$39,700 for both configurations, and break-even ticket prices rose to \$35–\$37, the highest of any clear operating day in the sample. This pattern was consistent across independent runs of both pad configurations, suggesting a constraint in the weather data for that date rather than stochastic noise in the simulation.

Clear-Day Network Performance. On the five fully clear operating days (February 5, 6, 8, 9, and 10), the network performed strongly across both configurations (see **Table 4**). The break-even ticket price of approximately \$31–\$32 across both configurations represents a substantial margin against the \$100 ticket price used in these runs. This gap reflects the relatively low variable cost per flight under autonomous operation and the high load factors achieved on established intercampus corridors. No congestion or loiter events were recorded in any run, indicating that neither configuration saturated pad capacity during the simulated period.

Two-Pad versus Three-Pad Configuration. The three-pad configuration consistently outperformed the two-pad configuration on clear days across all five metrics tracked. The daily flight advantage averaged approximately 51 additional flights per day, translating to roughly 135 additional passengers. Net daily profit was higher by an average of \$7,558, or approximately 13.9%. The break-even ticket price was nearly identical between configurations (\$31.64 versus \$32.02), indicating that the cost structure scales proportionally with throughput and the additional pad primarily functions as a capacity multiplier rather than a cost driver.

These results support the infrastructure design conclusion that the three-pad configuration provides meaningfully higher network throughput with no significant penalty in unit economics. Whether the incremental capital expenditure of adding a third pad is justified depends on projected demand. At the demand scale used in this simulation, the three-pad configuration operates below its capacity ceiling, suggesting that either configuration could absorb moderate demand growth before throughput becomes a limiting factor.

Break-Even Rate and Distance-Based Pricing. The simulation uses a flat fare of \$100 per passenger regardless of route distance. To assess whether this is economically appropriate across the network, the break-even variable cost per flight was derived by working backwards from the simulated clear-day average break-even ticket price of approximately \$32. Under autonomous operation, the variable cost per flight consists of two components: a fixed maintenance and battery reserve of approximately \$30.34 per flight, derived from the NASA eVTOL direct operating cost benchmark adjusted for pilotless operation, and an energy cost that scales with distance at approximately \$0.743 per mile, based on 90 kWh of charged energy per flight at California’s commercial electricity rate of \$0.231/kWh [9] and an assumed average efficiency of 28 miles per full charge cycle. Setting these two components equal to the break-even fare of \$32 yields an implied average leg distance of approximately 6.3 miles, which represents a worst-case short-hop interpretation of the break-even calculation.

In practice, the top routes in the simulation by departure frequency span a much wider range of distances, from 13 miles (KNUQ → KSQL) to over 58 miles (UCD → KLVK). Computing the haversine distance across the twenty most frequently flown corridors in the week-long dataset yields an average leg distance of approximately 27.5 miles. At this distance, the true variable cost per flight is approximately \$51, which is consistent with the simulated break-even results and confirms that the \$100 flat fare provides a comfortable margin on most routes while understating revenue potential on longer legs.

These results suggest that a distance-based pricing model would better reflect the actual cost structure of

the network. A rate of approximately \$6 per mile produces a fare of roughly \$38 on the shortest common legs (13 miles) and \$165 on a 27-mile average route, which is competitive with premium ground rideshare for short hops and appropriately higher for longer intercampus legs. For reference, longer routes such as UCD to UCB (approximately 53 miles) would yield a fare near \$320 at this rate, which is closer to the actual cost of premium ground transportation for that trip and more appropriate than the current flat \$100. **Table 5** summarizes the implied fares and margins at a \$6/mile rate across representative route distances.

The current flat-fare model is effectively underpriced on long routes and marginally overpriced on the shortest hops relative to break-even. Distance-based pricing would improve both revenue capture on longer corridors and passenger-side fairness on short ones, and is recommended for incorporation into future simulation runs.

Seven-Day Financial Summary. Over the full seven-day period including the February 4 storm day, the three-pad configuration produced a cumulative net profit of approximately \$331,740 and the two-pad configuration produced approximately \$293,080. Storm-day losses represented less than 1.6% of weekly revenue in both cases, consistent with the expectation that a single weather closure is financially manageable within the broader operating model. The full daily breakdown is provided in **Table 6**.

Site-Level Performance. Among the UC campus nodes, UCSC and UCB were consistently the highest-revenue sites in both configurations, reflecting their geographic position within the network and higher passenger volumes on connected corridors. UCM was the lowest-performing UC campus node in most runs, consistent with its relative geographic isolation from the airport cluster nodes. Among airport nodes, KNUQ and KSQL were the busiest by departure count, with the KNUQ $\rightarrow$ KSQL corridor appearing as the busiest or second-busiest route in the majority of runs. The KSNS $\leftrightarrow$ KCVH corridor was the most consistent high-frequency pairing across all clear days, generating 13 to 18 round-trip departures per day in nearly every run. No site recorded a loss on any clear operating day in either configuration, and no loiter or congestion events occurred in any run across the full week, indicating that the current fleet-to-pad ratio is well matched to simulated demand at this scale.

11 Conclusion

This proposal developed and evaluated a distributed Airlink concept that integrates vertiport infrastructure design, simulation based routing, safety logic, and economic modeling into a single decision support framework. Rather than treating vertiport construction as an isolated design problem, the work assessed how candidate infrastructure strategies support safe, scalable, and economically viable intercampus operations. Within that framework, parking structure retrofits emerged as a strong option for distributed campus nodes due to their reduced capital cost, avoidance of land acquisition, and compatibility with rooftop eVTOL operations.

The infrastructure analysis showed that existing parking structures can support compact vertiport layouts when evaluated using FAA informed sizing, spatial screening, and operational trade studies. In particular, the Lower Hearst case study demonstrated that a two pad configuration provides a highly integrated rooftop node, while a three pad configuration increases throughput at the cost of greater spatial and operational complexity. A simplified operational model further quantified this tradeoff, showing that throughput depends not only on pad count, but also on turnaround time, utilization, and queuing sensitivity.

At the system level, the simulation framework demonstrated that routing, weather assessment, airspace constraints, emergency planning, and sense-and-avoid logic can be integrated to generate meaningful operational outputs for the Airlink network. The results showed that the network remains economically promising under nominal conditions, while also capturing the performance impacts of adverse weather and operational disruptions. When coupled to the economic model, these outputs support the broader conclusion that simulation informed infrastructure planning can improve both cost effectiveness and deployment confidence.

Overall, this work supports the feasibility of a phased Airlink deployment strategy built around a combination of lower cost distributed retrofit nodes and larger regional hub connections. By linking infrastructure design, routing performance, safety constraints, and economic outcomes, the proposal advances a practical framework for scalable AAM implementation across the UC system. Future work should focus on refining in flight replanning, expanding demand calibration through survey or test scenarios, and further validating pad level operations under higher traffic scenarios.

A Figures and Renderings

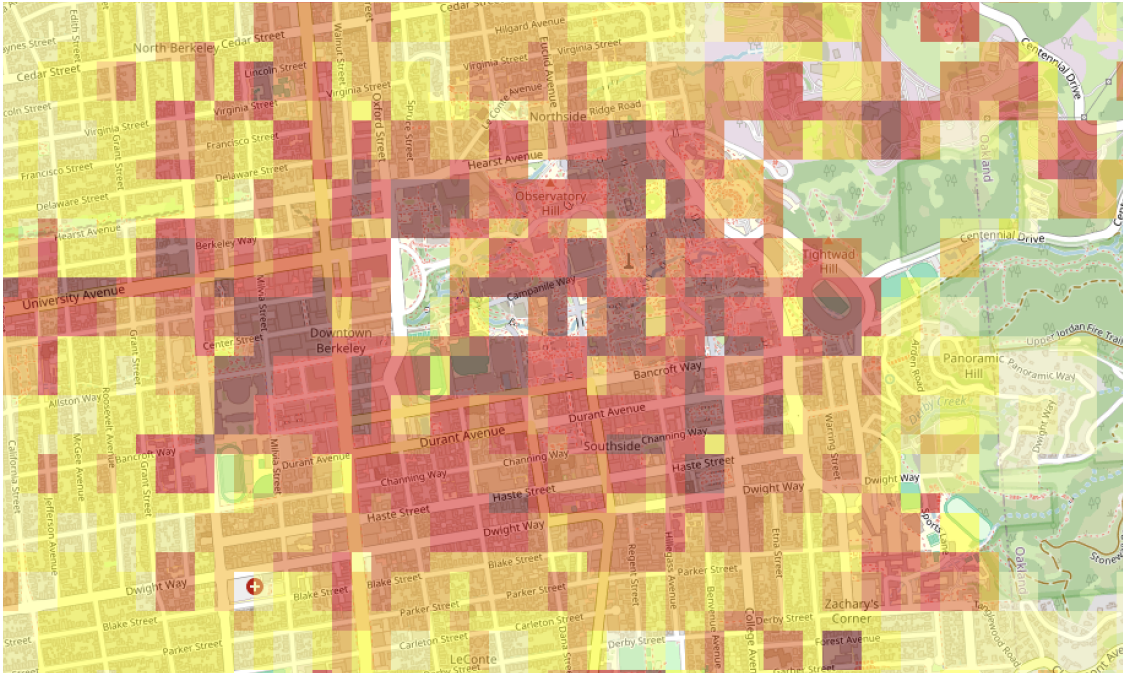


Figure 1: Daytime population density distribution near UC Berkeley. Higher density regions are shown in red, indicating increased exposure risk and higher demand concentration. These regions are incorporated into the routing model as weighted constraints to discourage unnecessary traversal.

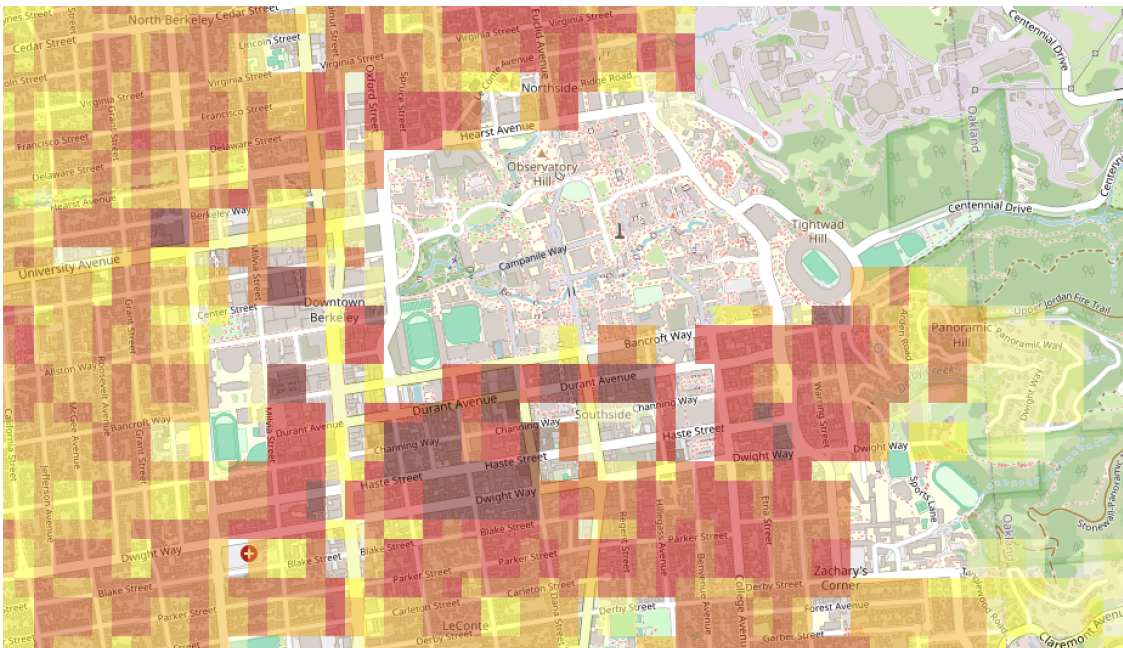


Figure 2: Nighttime population density distribution near UC Berkeley. The spatial shift in density reflects changes in activity patterns, which influence routing decisions and safety considerations within the Airlink simulation framework.

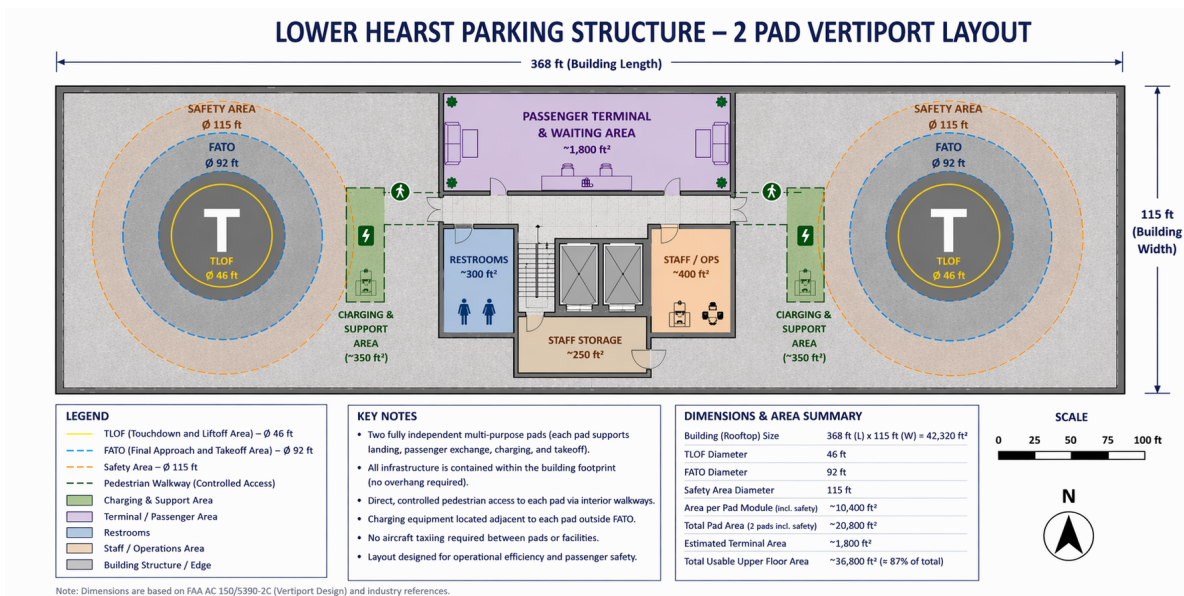


Figure 3: Conceptual two-pad vertiport layout for the Lower Hearst Parking Structure

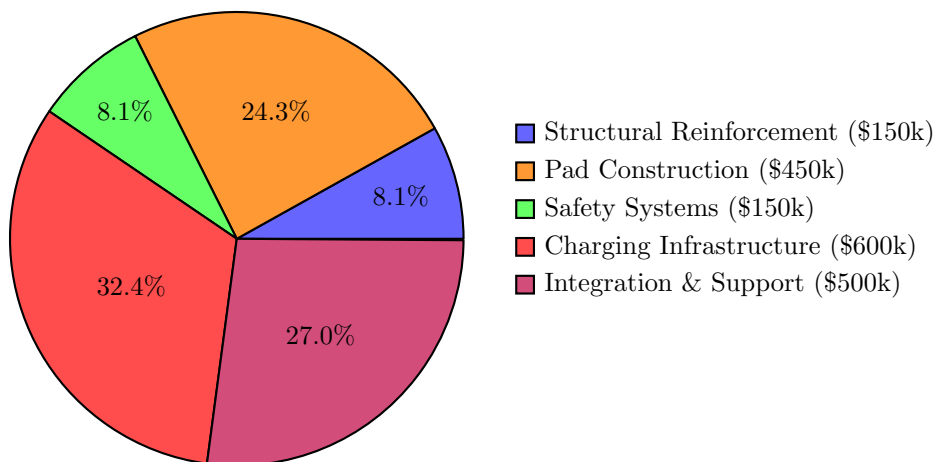


Figure 4: Capital cost distribution for the three-pad retrofit vertiport concept.

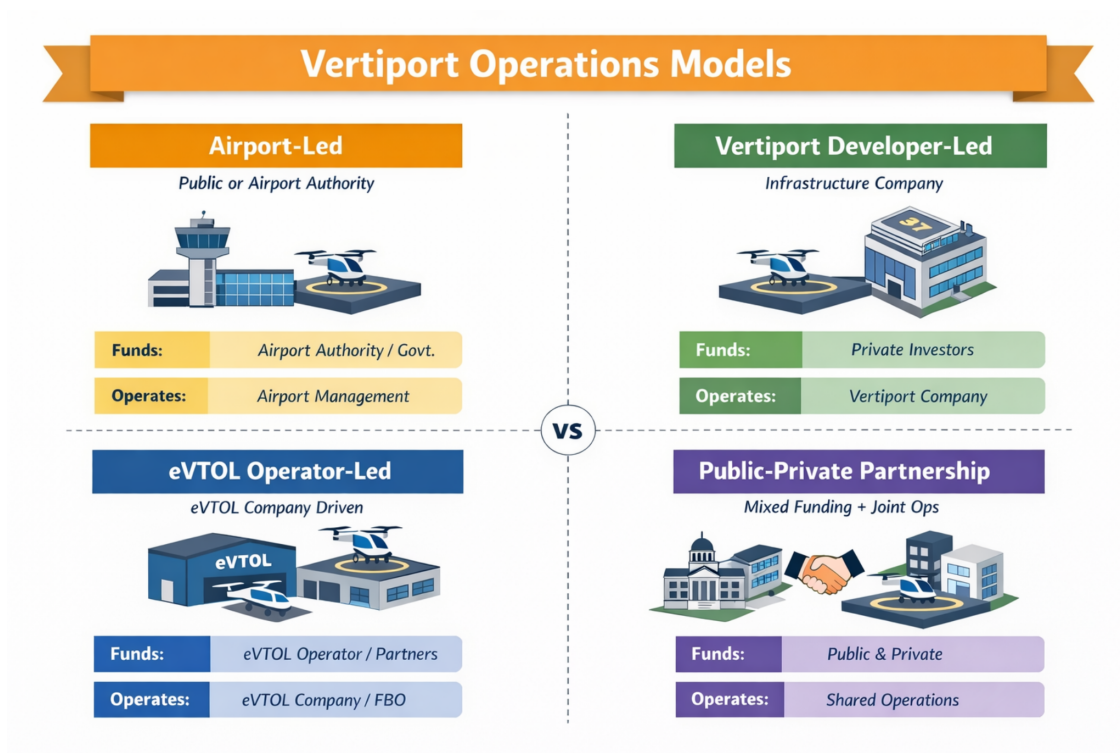


Figure 5: Representative vertiport ownership and operational models, including airport-led, developer-led, operator-led, and public-private partnership configurations.

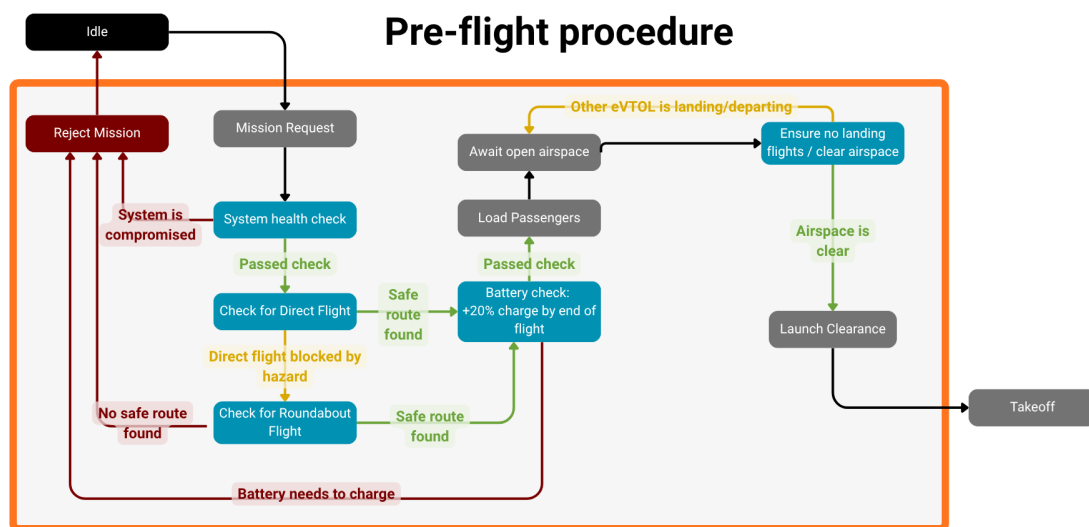


Figure 6: Pre-flight procedure for the Airlink system.

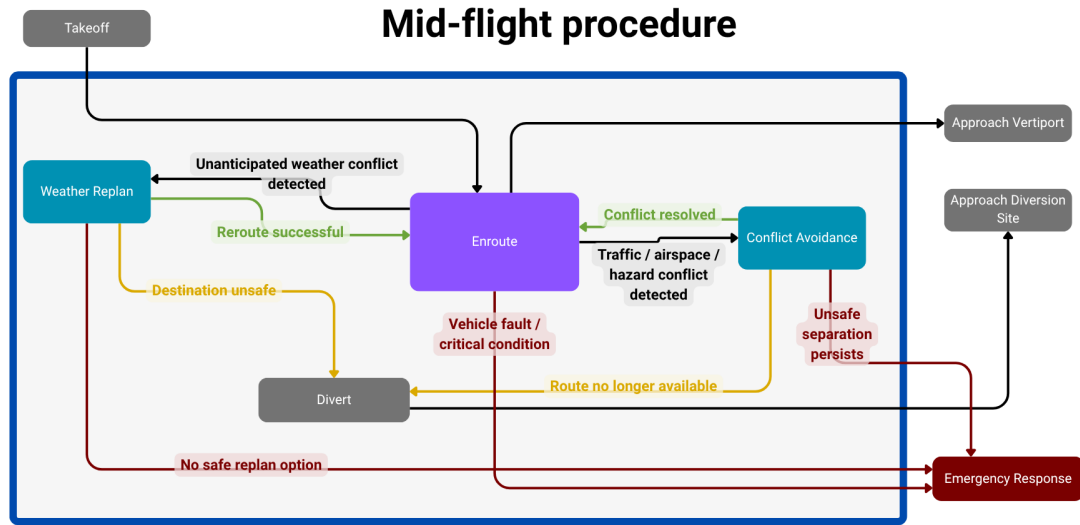


Figure 7: Mid-flight procedure illustrating enroute operations, including weather replanning, conflict avoidance, and diversion logic.

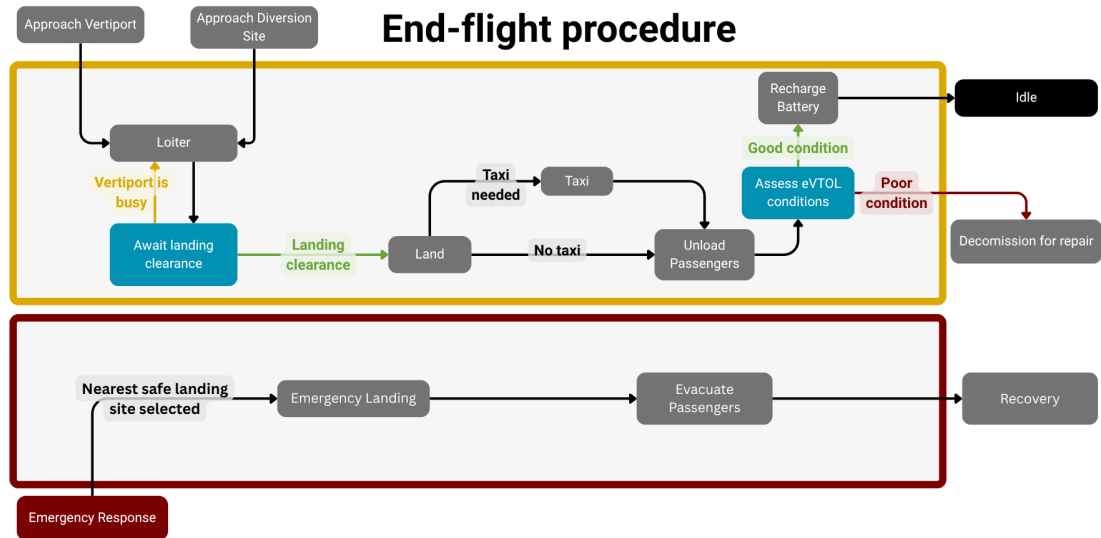


Figure 8: End-flight procedure showing vertiport approach, landing clearance, passenger unloading, and post-flight assessment. The process includes contingency handling for vertiport congestion and vehicle condition checks prior to re-entry into service or maintenance.

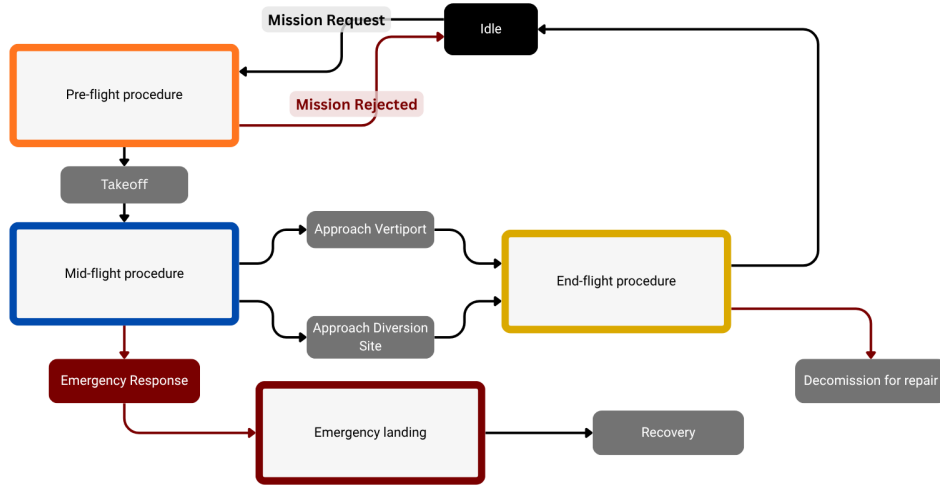


Figure 9: Integrated operational flow linking pre-flight, mid-flight, and end-flight procedures with emergency response pathways.

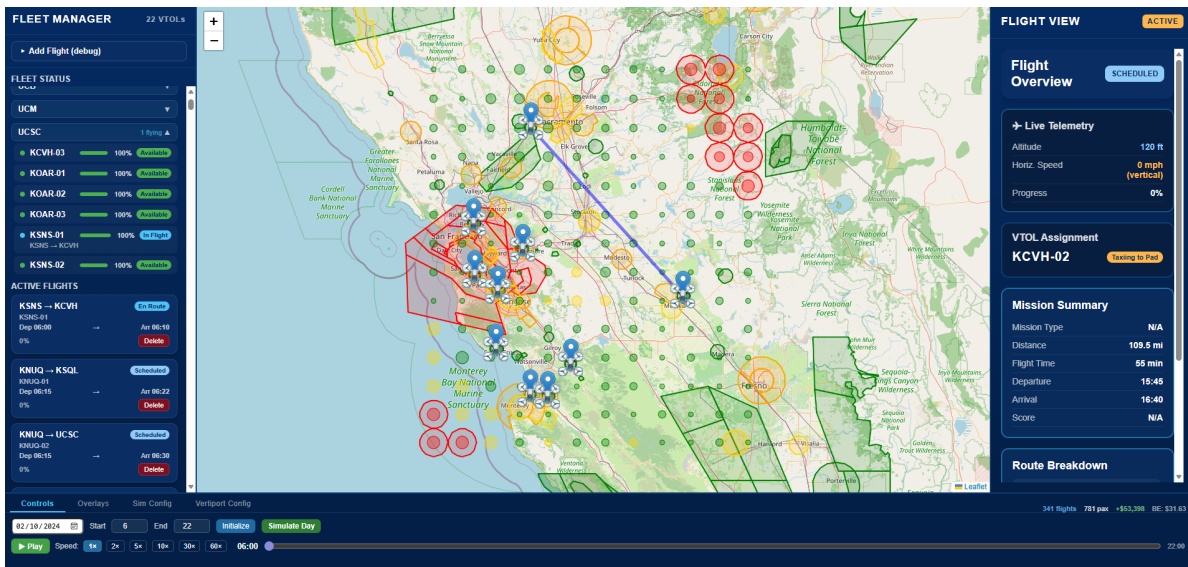


Figure 10: Simulation interface for the California Airlink system showing network-level operations across multiple vertiports. The visualization includes active routes, candidate vertiport locations, population density overlays, restricted airspace regions, and real-time mission status. The interface enables evaluation of routing behavior, flight scheduling, and system performance under varying operational and environmental conditions.

Table 1: AHP-Derived Site Selection Criterion Weights

Criterion	Weight
Safety Considerations (SAF)	0.35
Structural Capacity (STR)	0.24
Available Surface Area (AREA)	0.16
Accessibility (ACC)	0.10
Geometric Layout (GEO)	0.09
Electrical Infrastructure (ELEC)	0.06

Table 2: Top-Ranked Candidate Site by Campus

Campus	Top Candidate	Weighted Score
UC Berkeley	Lower Hearst Structure	4.20
UC Santa Cruz	Core West Parking Structure	3.94
UC Merced	North Bowl Lot	3.67
UC Davis	Gateway Parking Structure	4.26

Table 3: Estimated Annual Operating Cost Breakdown for a Three-Pad Retrofit Vertiport

Cost Category	Annual Cost (\$)	Notes
Staffing (Operations & Monitoring)	120,000	Assumes partial on-site + remote oversight
Maintenance & Inspection	60,000	Structural + pad + safety system upkeep
Utilities & Communications	30,000	Lighting, systems, baseline electrical
Insurance & Compliance	40,000	Regulatory, liability coverage
Security & Access Control	25,000	Cameras, access systems, monitoring
Lost Parking Revenue	50,000	Based on displaced spaces and utilization
Total Annual Infrastructure OPEX	325,000	

Table 4: Average clear-day simulation results by configuration

Metric	Three-pad (UC sites)	Two-pad (UC sites)
Flights completed	388	337
Passengers carried	906	771
Average load factor	2.3 pax/flight	2.3 pax/flight
Variable cost	\$22,804	\$19,602
Fixed daily cost	\$5,318	\$5,078
Net daily profit	\$61,878	\$54,320
Break-even ticket price	\$31.64	\$32.02

Table 5: Implied fares and variable cost margins at \$6/mile rate

Representative route	Distance (mi)	Fare at \$6/mi	Variable cost margin
KNUQ → KSQL (short)	13.2	\$79	\$44
KCVH → KSNS (medium)	19.3	\$116	\$65
UCB → KSQL (medium)	25.1	\$151	\$99
Network average leg	27.5	\$165	\$114
UCD → UCB (long)	53.4	\$320	\$249

Table 6: Seven-day simulation results summary

Date	Config	Flights	Pax	Cancelled	Net P/L
Feb 4 (storm)	3-pad	6	6	1,252	\$(5,125)
Feb 4 (storm)	2-pad	5	5	1,250	\$(4,913)
Feb 5	3-pad	372	850	160	\$57,558
Feb 5	2-pad	325	761	158	\$51,965
Feb 6	3-pad	386	873	178	\$59,417
Feb 6	2-pad	335	757	156	\$51,030
Feb 7 (partial weather)	3-pad	308	634	458	\$39,679
Feb 7 (partial weather)	2-pad	288	616	358	\$39,596
Feb 8	3-pad	402	930	140	\$64,076
Feb 8	2-pad	334	769	153	\$52,493
Feb 9	3-pad	387	874	176	\$59,455
Feb 9	2-pad	349	786	142	\$53,113
Feb 10	3-pad	393	901	149	\$61,683
Feb 10	2-pad	341	781	144	\$53,398
7-day total	3-pad	2,654	5,268	2,613	\$331,740
7-day total	2-pad	2,278	4,480	2,495	\$293,080

B Supplementary Materials

AHP Site Selection Analysis

A detailed description of the Analytic Hierarchy Process (AHP) used for site selection is provided in a supplementary document. This includes pairwise comparison matrices, weighting derivations, and campus-specific evaluation tables used to rank candidate parking structures.

Document Access:

<https://ucmerced.box.com/s/sfqcv72xlnhg572b82i4vhuyaqdhdxd1>

Economic Analysis Model

A detailed breakdown of the economic model used in this study is provided in a supplementary document. This includes capital expenditure (CAPEX) assumptions, operating cost (OPEX) estimates, cost-per-flight derivations, and sensitivity analyses for key parameters such as utilization, energy cost, and staffing levels. The model also includes scenario-based evaluations used to estimate system profitability and scalability under varying demand conditions.

Document Access:

<https://ucmerced.box.com/s/sfqcv72xlnhg572b82i4vhuyaqdhdxd1>

References

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- [4] CITRIS and the Banatao Institute, “Citris aviation prize,” 2025. Accessed April 2026.
- [5] California Airlink Team, “Ahp and decision matrix for candidate parking structures,” 2026. Internal project analysis document.
- [6] Federal Aviation Administration, “Engineering brief no. 105a: Vertiport design,” 2022.
- [7] Joby Aviation, “Technology,” 2025. Accessed April 2026.
- [8] National Renewable Energy Laboratory, “Electrification of aircraft: Charging infrastructure considerations,” 2021.
- [9] U.S. Energy Information Administration (EIA), “Electric power monthly: California commercial electricity price,” 2026. California commercial electricity price was 23.13 cents/kWh in January 2026. Used as baseline electricity cost for energy-per-flight calculations. Monthly statewide average; site-specific tariffs and demand charges may vary.